

KENDRIYA VIDYALAYA SANGATHAN
DELHI REGION
PRE-BOARD-1 2024-25
CLASS – 12
SUBJECT: PHYSICS

CLASS: XII

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M.M.-70

TIME:3 HOURS

General Instructions:

- (1) There are **33 questions** in all. All questions are compulsory.
- (2) This question paper has five sections: Section A, Section B, Section C, Section D and Section E.
- (3) All the sections are compulsory.
- (4) **Section A** contains **sixteen questions, twelve MCQs and four Assertion Reasoning based of 1 mark each; Section B** contains **five questions of two marks each; Section C** contains **seven questions of three marks each; Section D** contains **two case study-based questions of four marks each**, and **Section E** contains **three long answer questions of five marks each**.
- (5) There is no overall choice. However, an internal choice has been provided for one question in Section B, one in Section C, one in each CBQ in Section D, and all three questions in Section E. You have to attempt only one of the choices in such questions.
- (6) Use of calculators is not allowed.
- (7) You may use the following values of physical constants wherever necessary

- (a) $c = 3 \times 10^8 \text{ m/s}$
- (b) $m_e = 9.1 \times 10^{-31} \text{ kg}$
- (c) $m_p = 1.7 \times 10^{-27} \text{ kg}$
- (d) $e = 1.6 \times 10^{-19} \text{ C}$
- (e) $\mu_0 = 4\pi \times 10^{-7} \text{ T mA}^{-1}$
- (f) $h = 6.63 \times 10^{-34} \text{ J s}$
- (g) $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2\text{N}^{-1}\text{m}^{-2}$

Section -A

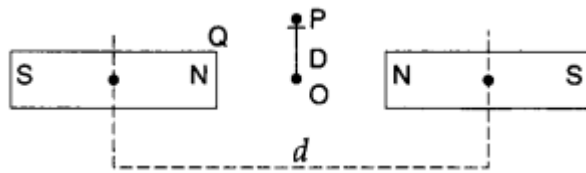
1. A charge 'q' is placed at the centre of a room having cubical shape. Find the electric flux through each wall of the room

(a) $\frac{q}{6\epsilon_0}$ (b) $\frac{q}{4\epsilon_0}$ (c) $\frac{q}{2\epsilon_0}$ (d) $\frac{q}{\epsilon_0}$

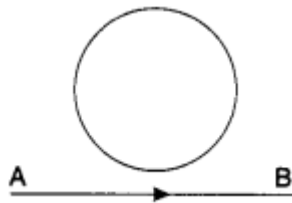
2. In a moving coil galvanometer, the deflection (Φ) on the scale by a pointer attached to the spring is:

(a) $\left(\frac{NA}{kB} \right) I$ (b) $\left(\frac{N}{kAB} \right) I$ (c) $\left(\frac{NAB}{k} \right) I$ (d) $\left(\frac{NAB}{kI} \right)$

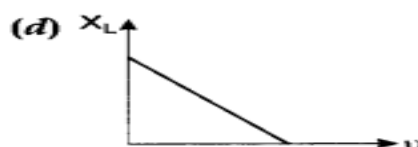
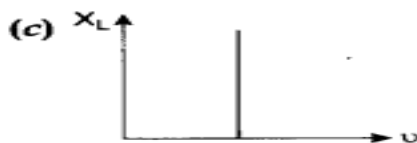
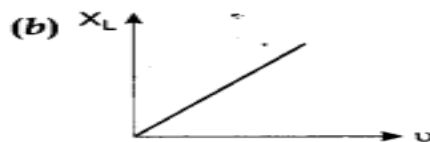
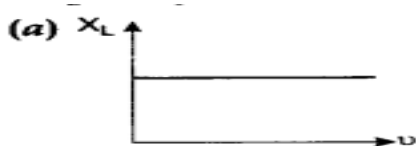
3. Two identical bar magnets are fixed with their centres at a distance d apart. A stationary charge Q is placed at P in between the gap of the two magnets at a distance D from the centre O as shown in the figure. The force on the charge Q is



- (a) zero
 (b) directed along OP
 (c) directed along PO
 (d) directed perpendicular to the plane of the paper
4. In the given figure, the current from A to B in the straight wire is decreasing. The direction of the induced current in the loop is A



- (a) clockwise
 (b) anticlockwise
 (c) changing
 (d) nothing can be said
5. Which of the following graphs represents the correct variation of inductive reactance X_L with frequency ν ?



6. If E and B denote electric and magnetic fields, respectively, which of the following is dimensionless?

(a) $\sqrt{\mu_0 \epsilon_0} \frac{E}{B}$

(b) $\mu_0 \epsilon_0 \frac{E}{B}$

(c) $\mu_0 \epsilon_0 \left(\frac{B}{E} \right)^2$

(d) $\frac{E}{\epsilon_0} \frac{\mu_0}{B}$

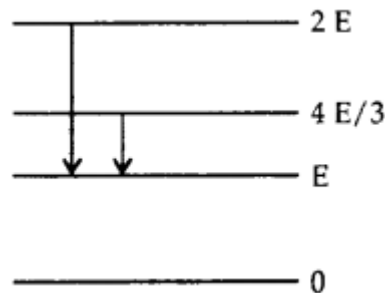
7. Two thin lenses of focal lengths 20 cm and – 40 cm are held in contact. If an object is placed at infinity, then the image will be formed by the combination at

- (a) 10 cm
- (b) 20 cm
- (c) 40 cm
- (d) infinity

8. The refractive index of glass is 1.5 for light waves of wavelength $\lambda = 6000 \text{ \AA}$ in vacuum. Its wavelength in glass is:

- (a) $2000 \text{ \AA}$
- (b) $4000 \text{ \AA}$
- (c) $1000 \text{ \AA}$
- (d) $3000 \text{ \AA}$

9. The adjoining figure indicates the energy levels of a certain atom; when the system moves from $2E$ to E level, a photon of wavelength λ is emitted. The wavelength of the photon produced during its transition from $4E/3$ to E is:



- (a) $\lambda/3$
- (b) $3\lambda/4$
- (c) $4\lambda/3$
- (d) 3λ

10. The binding energies per nucleon for a deuteron and an α -particle are x_1 and x_2 , respectively. The energy Q released in reaction ${}_1\text{H}^2 + {}_1\text{H}^2 \rightarrow {}_2\text{He}^4 + Q$ is:

- (a) $4(x_1 + x_2)$
- (b) $4(x_1 - x_2)$
- (c) $2(x_1 + x_2)$
- (d) $4(x_2 - x_1)$

11. The mobility of conduction electrons is greater than that of holes since electrons

- (a) are negatively charged.
- (b) are lighter
- (c) require smaller energy to move through the crystal lattice.
- (d) Undergo a smaller number of collisions.

12. In p-type semiconductor, the conduction current is due to

- (a) Holes
- (b) Atoms
- (c) Electrons
- (d) Protons

Directions: These questions consist of two statements, each printed as Assertion and Reason. While answering these questions, you must choose any of the following four responses.

(a) If both Assertion and Reason are correct and the Reason is a correct explanation of the Assertion.

(b) If both assertion and reason are correct, then reason is not a correct explanation of assertion.

(c) If the Assertion is correct but the Reason is incorrect.

(d) If both the Assertion and Reason are incorrect.

13. Assertion: If the distance between parallel plates of a capacitor is halved and the dielectric constant is three times, then the capacitance becomes 6 times.

Reason: The capacity of the capacitor does not depend upon the nature of the dielectric material.

14. Assertion: Induced e.m.f. will always occur whenever there is a change in magnetic flux.

Reason: Current is always induced whenever there is a change in magnetic flux.

15. Assertion: Though the light of a single frequency (monochromatic) is incident on a metal, the energies of emitted photoelectrons are different.

Reason: The energy of electrons emitted from inside the metal surface is lost in collision with the other atoms in the metal.

16. Assertion: The minimum and maximum wavelength ratio is $\frac{3}{4}$ in the Lyman series.

Reason: Lyman series constitute spectral lines corresponding to the transition from higher energy to the ground state of the hydrogen atom.

Section B

17. A potential difference of 6 V is applied across a conductor of length 0.12 m. Calculate the drift velocity of electrons if the electron mobility is $5.6 \times 10^{-6} \text{ m}^2 \text{ V}^{-1} \text{ s}^{-1}$.

18. Define magnetic field strength. Give its SI unit.

19. Will the current be induced if a square loop of conducting material is moved with a constant velocity fully inside a uniform magnetic field perpendicular to the field? Explain.

20. The de-Broglie wavelengths associated with an electron and proton are equal. Prove that the electron's kinetic energy is greater than that of the proton.

21. What happens to the width of the depletion layer of a p-n junction when it is

(i) forward biased?

(ii) reverse biased?

OR

Draw a labelled energy band diagram of a P-type and N-type semiconductor.

Section C

22. (a) What characteristic property of nuclear force explains the constancy of binding energy per nucleon (BE/A) in the range of mass number ' A ' lying $30 < A < 170$?

(b) Show that the density of nucleus over a wide range of nuclei is constant independent of mass number A .

23. a) Calculate the shortest wavelength of the spectral lines emitted in the Balmer series. (Rydberg constant = 10^7 m^{-1})

b) In which part of the electromagnetic spectrum all the wavelengths of the Balmer series fall?

OR

The total ground state energy of any hydrogen atom is -13.6 eV , then calculate:

(i) total energy, (ii) kinetic energy and (iii) potential energy in the first excited state?

24. (a) Draw the circuit diagram of a Full wave rectifier. Also, draw the input & Output waveforms.

(b) In half-wave rectification, what is the output frequency if the input frequency is 50 Hz ? What is the output frequency of a full-wave rectifier for the same input frequency?

25. Identify the constituent radiation of the Electromagnetic Spectrum, which

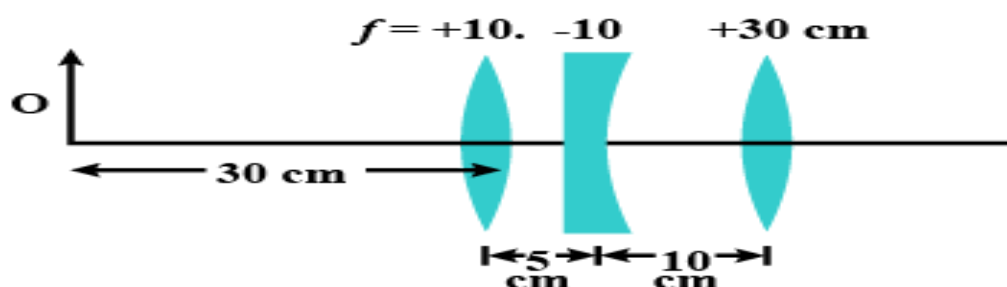
(i) is used to study crystal structure.

(ii) is absorbed by the Ozone layer in the atmosphere.

(iii) Mention one more application for each of this radiation

OR

Find the position of the image formed by the lens combination given in the figure.



26. State Huygens's postulates of wave theory. Using this, verify the laws of refraction.

27. Using Ampere's circuital law, obtain an expression for the magnetic field along the axis of a current carrying solenoid of length l and having N number of turns.

28. Using Gauss's law, derive an expression for electric field intensity due to an infinitely long, straight wire of linear charge density $\lambda \text{ C/m}$.

Section D

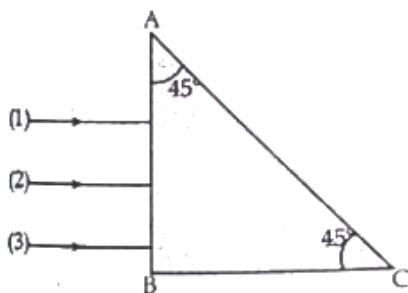
29. Total internal reflection is the phenomenon of light bouncing back into the same medium when light travels from one medium to another. For total internal reflection to take place, light has to travel from denser to rarer medium, and the angle of incidence has to be greater than the critical angle. Multiple internal reflections in diamonds, totally reflecting prisms, and mirages are some examples of total internal reflection.

- i) The refractive index of a material is $\sqrt{2}$. What should be the value of the incident angle so that the light undergoes total internal reflection?
- A) $\geq 45^\circ$ B) $\leq 45^\circ$
- C) $> 45^\circ$ D) $< 45^\circ$
- ii) For a ray of light entering from glass to water, which of the following statements is true for the critical angle
- A) Maximum for red colour and minimum for violet colour
- B) Maximum for violet colour and minimum for red colour
- C) is independent of colour of incident light
- D) Maximum for yellow colour
- iii) What will be the value of the angle of reflection when the angle of incidence is equal to the critical angle?
- A) 30° B) 45°
- B) 60° D) 90°

OR

What will be the speed of light in a medium whose critical angle is 30° ?

- A) 3×10^8 m/s B) 3×10^{-8} m/s
- C) 6×10^8 m/s D) 1.5×10^8 m/s
- iv) As shown in the figure, Three rays 1,2 & 3 of different colours fall normally on one of the sides of an isosceles right angle prism. The refractive index of the prism for these rays is 1.39, 1.47 and 1.52, respectively. Which of these waves will get internally reflected?



- A) 1 & 2
B) 2 & 3
C) 1 & 3
D) only 2

30. The electrolytic cell is a simple device that maintains a steady current in an electric circuit. A cell has two electrodes: positive (P) and negative (N). They are immersed in an electrolytic solution. Dipped in the solution, the electrodes exchange charges with the electrolyte. When a source of emf ε is connected to an external resistance R , the voltage V_{ext} across R is given by $V_{\text{ext}} = IR = \frac{\varepsilon}{R+r} R$, where 'r' is the internal resistance of the source.

i) A current of 2 A flows through a $2\ \Omega$ resistor when connected across a battery. The same battery supplies a current of 0.5 A when connected across a $9\ \Omega$ resistor. The internal resistance of the battery is

- | | |
|------------------|----------------|
| A) $1/3\ \Omega$ | B) $3\ \Omega$ |
| C) $0.5\ \Omega$ | D) $1\ \Omega$ |

ii) Maximum current can be drawn from the cell when

- | | |
|-----------------|------------|
| A) $R = \infty$ | B) $R = 0$ |
| C) $R = r$ | D) $r = 0$ |

OR

The maximum current that can be drawn from the cell is

- | | |
|----------------------------------|--------------------------------|
| A) $I = \frac{\varepsilon}{R+r}$ | B) $I = \frac{\varepsilon}{r}$ |
| C) $I = \frac{\varepsilon}{R}$ | D) $I = \frac{V}{R+r}$ |

iii) A battery has an emf of 4V and internal resistance r . When the battery is connected to an external resistance of $2\ \Omega$, a current of 1 A flows through the circuit. How much current will flow if the terminals are directly connected?

- | | |
|-------------|--------|
| A) 0 A | B) 1 A |
| C) infinite | D) 2 A |

iv) For an open circuit, which of the following statements is true

- | | |
|----------------------|-------------------------|
| A) $\varepsilon = V$ | B) $\varepsilon > V$ |
| C) $\varepsilon < V$ | D) $\varepsilon \leq V$ |

Section E

31. (a) A resistor of $200\ \Omega$ and a capacitor of $15.0\ \text{mF}$ are connected in series to a 220 V, 50 Hz AC source.

- Calculate the current in the circuit;
- Calculate the voltage (rms) across the resistor and the capacitor.
- Is the algebraic sum of these voltages more than the source voltage? If yes, resolve the paradox.

(b) In a series LR circuit, the inductive reactance (X_L) is numerically equal to resistance (R) of a resistor used. (i.e. $X_L = R$) The power factor of the circuit is P_1 . When a capacitor with capacitance (C) such that $X_L = X_C$ is put in the circuit, the power factor becomes P_2 . Calculate P_1/P_2 .

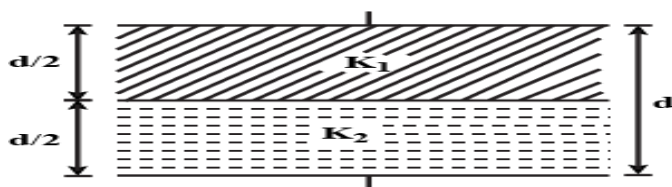
OR

Explain, with the help of a diagram, the principle and the workings of a step-up transformer, and

obtain the expression for the transformer equation. $\frac{i_p}{i_s} = \frac{V_s}{V_p} = \frac{N_s}{N_p}$, where the symbols have their usual meanings.

32. (a) What is a capacitor? Deduce an expression for the capacitance of a parallel plate capacitor fully filled with dielectric medium between the plates.

(b) A parallel plate capacitor of plate area A and separation d is filled with dielectrics of dielectric constant K_1 and K_2 , as shown in the figure. Find the net capacitance of the capacitor.



OR

A parallel plate capacitor is charged by a battery. After some time, the battery is disconnected, and a dielectric slab of dielectric constant K is inserted between the plates. What change will take place in the following :

- (i) Charge on the plates
- (ii) the capacitance of the capacitor
- (iii) potential difference between the plates
- (iv) electric field intensity between the plates
- (v) the energy stored in the capacitor? Justify your answer in each case.

33. (A) Draw a ray diagram to form an image using a compound microscope. Write the expression for its magnifying power, when the image is formed at infinity.

(B) A compound microscope uses an objective lens of a focal length of 4 cm and an eyepiece lens of a focal length of 10 cm. An object is placed at 6 cm from the objective lens. Calculate the magnifying power of the compound microscope.

OR

In Young's double-slit experiment, the two slits are 0.15 mm apart. The light emitted by the source has a wavelength of 450 nm. The screen is 2 m away from the slits.

- (i) Find the distance of the second bright fringe and the third dark fringe from the central maximum.

(ii) Find the fringe width.

(iii) What will happen to the fringe width if the whole setup is immersed in water with a refractive index of $4/3$?
